

Research Paper

Endemic Dental and Skeletal Fluorosis: Effects of High Groundwater Fluoride in some South Indian Villages

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Groundwater is an essential and vital component of our life support system. The groundwater resources are being utilized for drinking, irrigation and industrial purposes. There is growing concern of deterioration of groundwater quality due to geogenic and anthropogenic activities. A groundwater quality survey has been carried out in Uravakonda, Anantapur district, Andhra Pradesh. Most of the chemical concentrations collected from the study area do not comply with the water quality standards. Fluoride concentrations in groundwater samples of the study area varied between 0.5 mg/l to 4.5 mg/l. Majority of the samples are exceeding the permissible limit of fluoride 1.5 mg/l (WHO 2004). Probable sources of fluoride are weathering and leaching of fluoride bearing minerals under the alkaline environment. Further the concentration of fluoride is aided by high rate of evaporation, longer residence time of waters in the aquifer zone, intensive and long term irrigation. Dental and skeletal fluorosis and deformation of bones in children as well as adults were observed in the study area indicating the health impact of fluoride in groundwater.

Key Words: Quality, Fluoride, Anantapur, A.P

Introduction:

Water is essential natural resource for sustaining life and environment which we have always thought to be available in abundance and free gift of nature. Groundwater forms a major source of drinking water in urban as well as in rural areas. Groundwater has the properties of dissolving and carrying in solution, a variety of chemical and other materials. More than 90% of the rural population uses groundwater for domestic purposes. Major problems are being faced by the country due to the presence of excess fluoride, arsenic and nitrate in groundwater in certain parts of country. At present twenty nine countries are reported to be affected with fluorosis, the fluoride related disease. Fluoride problems are wide spread in nine states covering almost the entire country. In India about 62 million people in 17 states including Andhra Pradesh are affected with dental, skeletal and or non-skeletal fluorosis (Susheela, 1999). Fluoride ion in drinking water is known for both beneficial and

detrimental effects on health. It is essential for normal mineralization of bones and formation of dental enamel with presence in small quantity (Chouhan and Flora 2010). However, when consumed in higher doses ($>1.5\text{mg/l}$), it leads to dental fluorosis or mottled enamel and excessively high concentration ($>3.0\text{mg/l}$) of fluoride may lead to skeletal fluorosis (Raju et al., 2009). The amount of fluoride present naturally in groundwater is governed by climate, pH, composition of the host rock and hydrogeology (Gupta et al 2006, Raju et al 2009). Some anthropogenic activities are also contributed to cause an increase in fluoride concentration in groundwater such as use of phosphatic fertilizers, pesticides, sewage and sludges, depletion of groundwater table etc (Ramanaiah et al 2006). The degree of fluorosis has been related to the concentration of fluoride in drinking water and period of consumption. The present paper reports on the prevalence of fluoride concentration in ground waters of uravakonda revenue mandal of Anantapur district, Andhra Pradesh; where the people rely on groundwater as potable water source.

Some major inorganic constituents were also

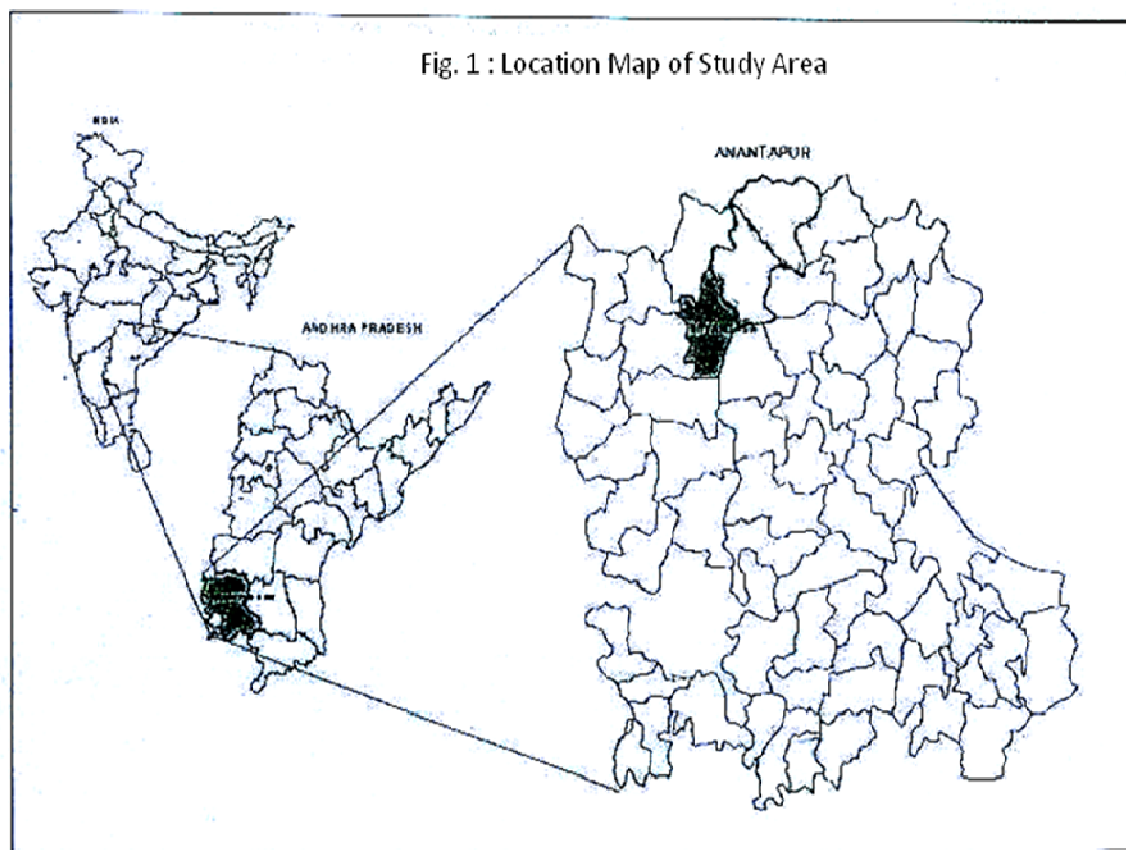


Figure 1: Location Map of Study Area

Determined to ascertain the water quality.

Study Area

The study area lies between longitudes $77^{\circ} 15' - 77^{\circ} 20'$ east and latitudes $15^{\circ} 0' - 14^{\circ} 50'$ north and falls in the Survey of India Toposheet No: 57 F/5 as shown in the figure:1 Sample locations are shown in figure -2. The study area comprises pink Granites, schists, composite gneisses of Dharwar intruded by a few pegmatite dykes and numerous dolerite dykes and the possible diamondiferous volcanic pipes (Anonymous 1993; Ramam and Murthy 1997). The major geomorphic units of the study area are Denudational Hills, Dissected pedimonts, Pediplain, and Alluvium. The study area lying off the coast does not enjoy the full benefit of the north.

Methodology

Twenty Five groundwater samples were collected from 20 different locations of Uravakonda area, Anantapur

District, Andhra Pradesh during April and November 2010. The sample location map of the study area is shown in Figure-1. The samples were collected from bore wells which are extensively used for drinking and other domestic purposes. The collected groundwater samples were analyzed for pH, electrical conductivity (EC), using pH and EC meters; Total dissolved solids (TDS), which were computed by multiplying the EC with a factor (0.55 to 0.75) that depend on the relative concentrations of ions; Total alkalinity (TA) as CaCO_3 and bicarbonate HCO_3^- were estimated by titrating with HCl; Total hardness (TH) as CaCO_3 and Calcium (Ca^{2+}) were analyzed titrimetrically, using standard EDTA; Magnesium (Mg^{2+}) was calculated on the basis of the difference in the concentration between TH and Ca^{2+} ; Sodium (Na^+) and Potassium (K^+) were measured by flame photometry; chloride (Cl^-) was estimated by standard AgNO_3 titration; sulphate (SO_4^{2-}), nitrate (NO_3^-) was analysed by spectrophotometer; fluoride (F) concentrations in mg/l except pH (units) and EC in microseimens/cm The method of collection and analysis of water samples followed are essentially the same as given by APHA (1992).

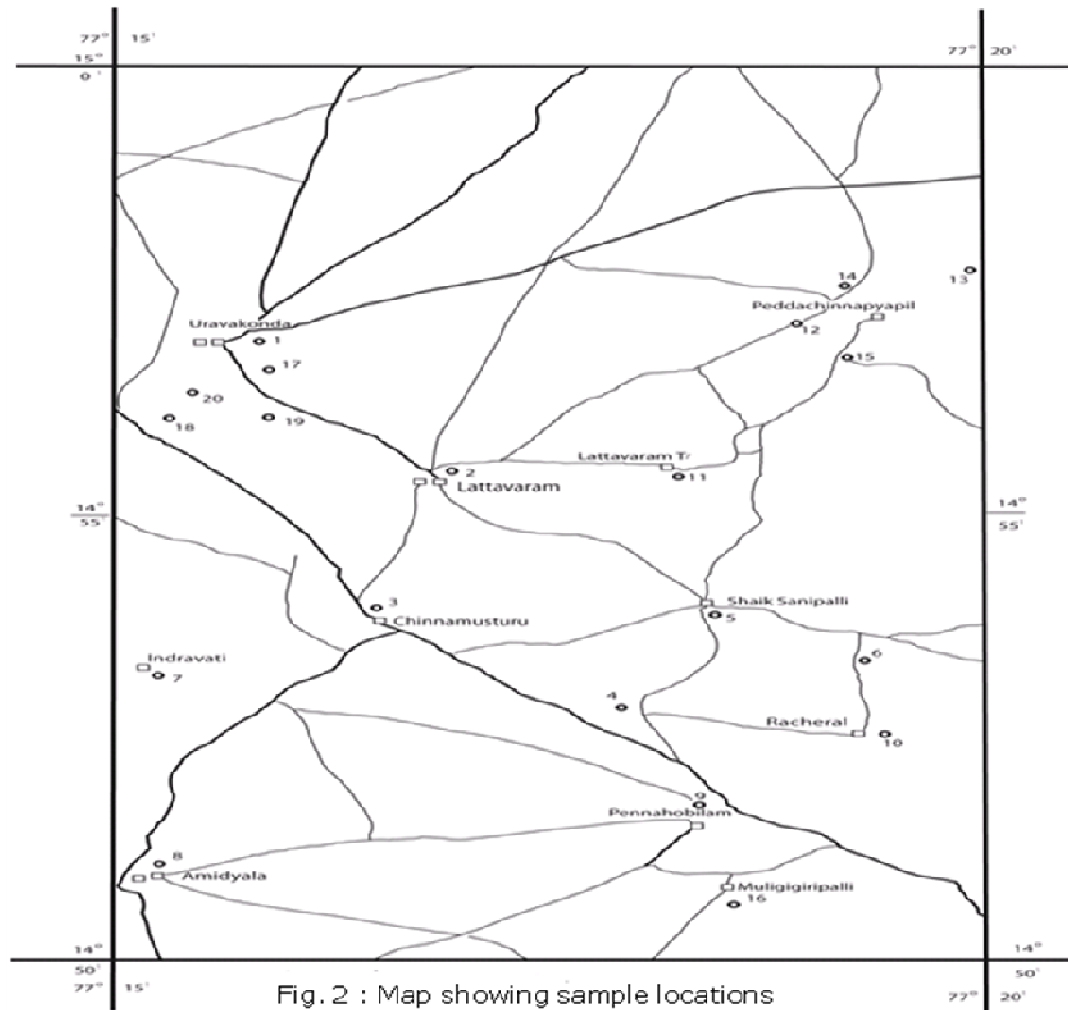


Figure 2: Map showing sample locations



Figure 3: Dental Fluorosis from Bapatkunta village of Mudigubba mandal of Anantapur District,

MANIFESTATION OF SKELETAL FLUOROSIS IN RALLA ANANTAPUR VILLAGE, ANANTAPUR DISTRICT, ANDHRA PRADESH



Figure 4: Skeletal fluorosis from Ralla anantahapuram village, Anantapur District, Andhra Pradesh

Results and Discussion:

The result of chemical analysis of fluoride concentration is presented in table 1. Fluoride concentrations in the study area varied between 0.5 to 4.5 mg/l. The analytical data revealed that the groundwater in the study areas is alkaline. Total dissolved solids values of all samples exceed highest desirable limit of 500 mg/l (ISI, 1983). Total Hardness values of all samples exceed the highest desirable limit of 300 mg/l. Water hardness is primarily due to the result of interaction between water and the geological formations (Angino, 1983). The fluoride concentration of the groundwater in the study area ranges from 1.7 to 6.8 mg/l with an average of 4.2 mg/l during pre-monsoon period and from 1.5 to 6.8 mg/l with an average of 3.3 mg/l during post-monsoon period. The permissible limit of fluoride in drinking water is 1.5 mg/l (ISI, 1983). It is observed that nearly 95% of the groundwater of the area exceeds the desirable limits during pre monsoon and post-monsoon periods. Fluoride concentration in natural waters depends on various factors such as temperature, pH, solubility of fluorine-bearing minerals, anion exchange capacity of aquifer materials (OH^- for F^-) and the nature of geologic formations drained by water and contact time of water with a particular formation. Minerals which have the greatest effect on the hydrogeochemistry of fluoride are fluorite, apatite, mica, amphiboles certain clays and villiamite. The Anantapur district of Andhra Pradesh

State, India, is a typical region showing endemic fluorosis caused by drinking water. Geological formation is found to be a basic cause for the higher concentration of fluoride in most of the sampling points.

CONCLUSIONS

The present water availability situation in Uravakonda, Anantapur District, Andhra Pradesh, is under great threat. Most of the chemical concentrations collected from the study area do not comply with the water quality standards. Geological formation is found to be a basic cause for the higher concentration of fluoride in most of the sampling points. Alkaline environment is the dominant controlling factor for leaching of fluoride from the source material in the groundwater. The high fluoride content in the groundwater of this area has affected villagers in the form of primary level of fluorosis resulted in stained and darkened tooth enamel. Several reports on dental and skeletal manifestations of fluorosis are also reported in the study area, which shows that the population of the study area is at higher risk due to excessive fluoride intake.

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Table 1: Analytical data of the chemical analyses of groundwater in Study area (in mg/l)

S.No	pH	EC μS/cm	TDS mg/l	TH mg/l	Na ⁺ mg/l	K ⁺ mg/l	Ca ⁺² mg/l	Mg ⁺ mg/l	HCO ₃ ⁻ mg/l	SO ₄ ⁻ mg/l	Cl ⁻ mg/l	F ⁻ mg/l	NO ₃ ⁻ mg/l
1	7.4	420	1040	678	720	60	170	70	170	240	140	4.5	58
2	7.6	620	1050	650	780	68	180	60	160	210	170	3.15	55
3	7.3	510	2470	655	800	70	200	50	140	200	190	2.49	60
4	7.5	900	2450	346	850	72	70	58	150	140	170	2.65	68
5	8.3	850	2200	413	730	65	90	60	180	144	80	4.37	90
6	8.2	700	1800	496	840	40	110	62	210	160	82	1.5	92
7	8.3	350	1620	246	850	48	120	58	220	210	190	4.06	70
8	8.3	580	3020	505	780	50	130	50	230	220	95	3.93	30
9	8.3	310	1050	538	760	58	140	55	200	230	98	3.06	62
10	8.4	340	3050	596	820	60	150	64	210	280	110	0.5	45
11	7.2	360	2020	662	810	72	160	72	180	300	120	1.89	48
12	7.0	600	2310	728	840	50	200	70	160	95	130	5.51	56
13	8.0	680	1020	630	680	54	180	60	170	90	134	1.5	70
14	8.3	650	1080	621	700	80	190	68	130	98	220	4.03	72
15	8.4	460	1000	637	820	85	170	76	120	100	210	1.51	56
16	8.5	920	880	494	810	40	150	80	90	120	190	0.54	48
17	8.6	900	850	478	520	42	80	82	80	150	94	0.65	98
18	7.9	880	900	490	430	36	70	70	75	158	96	1.86	78
19	8.2	620	920	596	410	50	90	60	95	140	110	1.57	65
20	8.3	710	980	530	380	60	150	50	110	210	124	3.93	60
21	8.0	760	1000	539	350	58	140	45	140	95	128	0.74	20
22	8.6	630	1600	580	360	70	160	40	160	80	330	3.81	70
23	7.5	680	2000	639	310	72	180	42	150	210	410	3.47	100
24	7.6	700	2600	540	280	78	200	30	130	220	210	1.49	46
25	8.2	720	1500	398	450	70	120	38	138	98	180	2.28	56

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